

Technical note

Tritium: its generation and pathways to the environment at CANDU 6 generating stations

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Abstract

Tritium production rates and its pathways in CANDU 6 generating stations are studied in this paper. Tritium is generated primarily by the $D(n, \gamma)T$ reaction in the reactor core. The major sources of tritium are moderator and coolant heavy water which are exposed to high neutron flux levels during reactor operation. A small amount of tritiated heavy water escape from heavy water filled systems is inevitable. Most of the tritiated heavy water which escapes is recovered through well-equipped heavy water recovery and collection systems. The vaporized tritiated heavy water is controlled by the D_2O vapour recovery and ventilation systems. Similarly, the liquid tritiated heavy water is collected by portable D_2O recovery tools and by an active drainage system. The remaining small amount escapes as tritiated heavy water and is lost to the environment in airborne and/or waterborne form.

1. Introduction

Natural uranium is used as nuclear fuel and heavy water is used as moderator and coolant in CANDU 6 generating stations (GS). The moderator and coolant heavy water are physically separated as shown in Fig. 1. There are several tritium generation reactions: $D(n, \gamma)T$, $^{10}B(n, 2\alpha)T$, $^7Li(n, \alpha)T$, $^3He(n, p)T$ and $^{235}U(n, f)T$. Of these, the $D(n, \gamma)T$ reaction is the major tritium source in CANDU 6 reactors.

Tritium exists in molecular form, i.e. DTO or T_2O (see Fig. 2). Most tritiated heavy water (THW) in the reactor core exists in DTO form, since the probability of activating two deuterium atoms in the same molecule in the reactor core is

extremely low compared with the probability of activating a single deuterium atom.

Fig. 3 shows the heavy water handling systems. Mixing between the primary coolant and moderator heavy water is kept to a minimum. During normal operation, the THW is contained in the closed coolant (sometimes called the heat transport) and moderator systems. However, leakage from the heavy water filled systems is inevitable. Most of the leakage occurs from the hot, pressurized coolant system.

The balance equation of THW is as follows:

$$\text{Leakage} = \text{Recovery} + \text{Loss}$$

where Leakage (or Escape) is the amount of heavy water leaking from the intended contain-

ment, Recovery is the amount of heavy water which is recovered and Loss is the amount of heavy water which is released to the environment.

Most of the THW leakage is recovered via the heavy water vapour recovery system or the active drainage system. However, a small amount of THW leaking from the systems and components is lost through the main stack or via the condenser cooling water canal into the environment.

It is important to minimize leakage from the equipment and components of the THW filled systems, so that THW losses are reduced and, consequently, the tritium hazard to station personnel and members of the public is minimized. Heavy water is an expensive material and its cost drives much of the tritium control features of the plant. Stopping the leaks of heavy water is fundamental to tritium control.

2. Sources of tritium

Tritium at a CANDU 6 GS is generated in the thermal neutron flux field. A CANDU 6 GS has the following sources of tritium: moderator system, coolant system, shield cooling system and liquid zone control system. The tritium activity, total tritium inventory and major tritium production reactions for each system at Wolsong 1 are given in Table 1.

3. Tritium activity in the reactor core

The tritium activity in the moderator system and coolant system heavy water as a function of the station operating time can be calculated assuming that there is no mixing between coolant and moderator heavy water and there is no tritium displacement programme.

The intent of a tritium displacement programme is to reduce the tritium activity in the moderator and coolant heavy water by replacing high activity heavy water with virgin heavy water. A tritium displacement programme contributes to reducing radiation exposure for the station personnel.

The tritium specific activity in a system is given by

$$A_t = A_0 e^{-(\lambda + L/M)t} + \frac{MR + LN_r}{M + L/\lambda} [1 - e^{-(\lambda + L/M)t}]$$

where A_t is the tritium activity at time t (Bq kg-D₂O⁻¹), A_0 is the initial tritium activity (Bq kg-D₂O⁻¹), λ is the tritium decay constant (s⁻¹), L is the D₂O replacement rate which is equal to the D₂O loss rate (kg s⁻¹), M is the total D₂O circulating mass in the system (kg), N_r is the tritium concentration in replacement D₂O (T kg-D₂O⁻¹), R is the reaction rate (CN_DΦσc) (T atoms kg-D₂O⁻¹ s⁻¹), C is the mass fraction of D₂O in the core, N_D is the deuterium concentration in the system (D atoms kg-D₂O⁻¹), Φ is the thermal neutron flux at full power (n cm⁻² s⁻¹), σ is the deuterium thermal neutron capture cross-section (cm²), c is the station capacity factor and t is the station operating time (s).

The theoretical equilibrium tritium activity in the moderator, based on operation at 84% capacity factor, is 3.26×10^{12} Bq kg-D₂O⁻¹ and that of

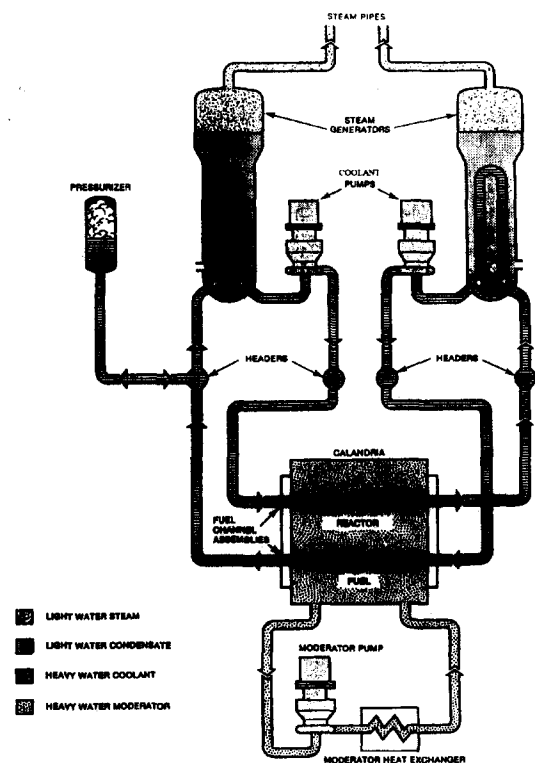


Fig. 1. The moderator and coolant systems for CANDU 6 GS.

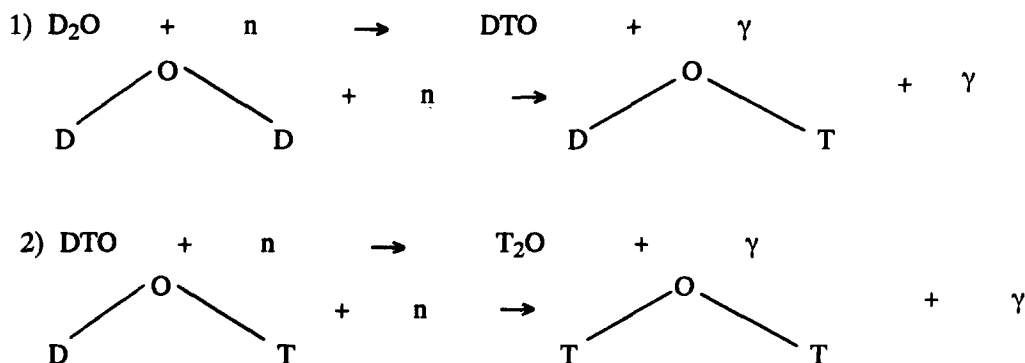


Fig. 2. The generation reactions of the molecular form of tritium in the reactor core at CANDU 6 GS.

the coolant is $7.21 \times 10^{10} \text{ Bq kg-D}_2\text{O}^{-1}$. The historical trend of tritium activity at Wolsong 1 is given in Table 2.

4. Leakage of tritium

Some THW may leak from systems through routine operations of the station via pump seals and valve packings in a controlled fashion. These losses are called “chronic” leakages. Such operations include chemical sampling of the THW systems, deuteration and dedeuteration, transfer of heavy water and draining of equipment for maintenance.

THW may also leak during acute events. This additional leakage of THW is termed “acute” leakage. These acute events include spills, steam generator or heat exchanger tube leaks and delayed neutron monitoring system tube leaks.

According to the operating data from 1983 to 1990, the average leakage rate for the CANDU

6 GS is about $23.6 \text{ Mg year}^{-1}$, 81% of which is chronic and 19% is acute.

The selection of components, their leak tightness and the actual design details within a system are interrelated and thus they all affect the overall leak tightness of a system. Various types of mechanical joints, pipe fittings, such as swagelok couplings, and threaded connections make only a small contribution, whereas flanges and gate valves make larger contributions per component. Shafts on valves and on high speed rotating devices are also a major contribution to the total leakage. There is therefore a preference for welded joints in heavy water filled systems and a preference to use valves with very low leak rates. Hence valves whose stems are bellows sealed or which have live loaded stem packings are used wherever possible.

As the station becomes “mature”, there may be acute leakages through tubes due to fretting and wear. Corrosion on tubes or thin components, vibration at/of mechanical joints and inadvertent operator actions may also cause acute leakage.

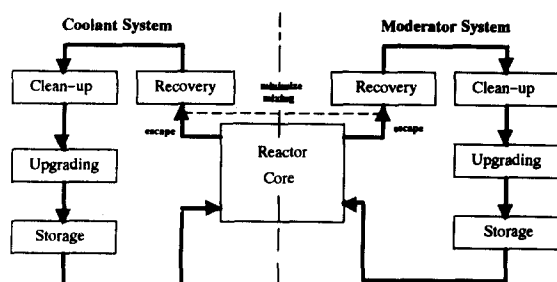


Fig. 3. Heavy water handling systems at CANDU 6 GS.

5. Escape pathways of tritium

THW leakage occurs as vapour or liquid. Thus the station systems are designed to collect and recover the THW as quickly as possible to reduce downgrading and to protect against radiation hazard to station personnel.

Table 1
Sources of tritium at Wolsong 1^a (as of end of 1994)

System	Tritium activity (Bq kg ⁻¹) ^b	Tritium inventory (TBq) ^c	Tritium production reaction(s)
Coolant	5.7×10^{10}	1.16×10^4	D(n, γ)T, ^7Li (n, α)T, ^{235}U (n, f)T
Moderator	1.43×10^{12}	3.76×10^5	D(n, γ)T, ^{10}B (n, 2α)T, ^3He (n, p)T
Spent fuel bay (SFB)	8.70×10^6	23.74	Leaked D ₂ O during refuelling ingress into SFB system
Shield cooling	5.54×10^7	31.41	D(n, γ)T
Liquid zone control	7.52×10^7	0.64	D(n, γ)T

^a The first reactor critical was reached on 21 November 1982. ^b The unit of tritium activity is Bq kg-D₂O⁻¹ or Bq kg-H₂O⁻¹ depending on its working fluid. ^c 1 TBq represents 1×10^{12} Bq. ⁷ Li is present in the coolant as a result of chemistry control. ¹⁰ B is a soluble poison in the moderator. ³ He is the decay product of tritium.

Escaped heavy water is in contact with the surrounding atmosphere in the reactor building (which contains water vapour), dirt and oil and is recovered, by the D₂O vapour recovery system and the active drainage system, as downgraded heavy water. Downgraded heavy water causes extra expenditure for the required upgrading. The downgraded heavy water is cleaned up and upgraded through the process given in Fig. 3.

Table 2
Historical trend of tritium activity at Wolsong 1

Year	Capacity factor (%)	Tritium activity (Bq kg-D ₂ O ⁻¹) ^a	
		Moderator	Coolant
1983	62	11.1×10^{10}	0.4×10^{10}
1984	67	23.2×10^{10}	0.6×10^{10}
1985	94	40.4×10^{10}	1.1×10^{10}
1986	80	60.5×10^{10}	1.9×10^{10}
1987	93	67.3×10^{10}	2.0×10^{10}
1988	79	79.0×10^{10}	2.5×10^{10}
1989	91	102.6×10^{10}	3.1×10^{10}
1990	86	125.5×10^{10}	4.4×10^{10}
1991	91	127.5×10^{10}	4.8×10^{10}
1992	87	131.0×10^{10}	5.1×10^{10}
1993	100	138.4×10^{10}	5.4×10^{10}
1994	83	142.5×10^{10}	5.7×10^{10}

^a The activity was measured at the end of each year. Wolsong 1 has not applied any tritium displacement programme to date.

To reduce intermixing between the coolant heavy water (low tritium) and the moderator heavy water (high tritium), the reactor building is divided into four dried areas, each with its own dedicated D₂O vapour recovery drier unit, and a ventilated area (see Fig. 4), i.e. fuelling machine vaults, fuelling machine service rooms and accessible areas, moderator equipment enclosure and steam generator room.

The interzonal flows maintain separation between areas with low levels of tritium contamination and areas with higher levels of tritium contamination. In particular, the interzonal flows between the ventilated area and the dried area ensure that there is little THW vapour released via the ventilation system to the stack.

For Wolsong 2, 3 and 4 (improved CANDU 6 GS currently under construction near Kyungju, South Korea), a fifth area consisting of the moderator purification valve gallery and heavy water management area in the service building, is connected (see Fig. 4) to the same D₂O vapour recovery system as the moderator equipment enclosure. In addition, various design improvements, including monitoring systems, have been introduced into the potential tritium release pathways to reduce tritium releases (Park and Aydogdu, 1993).

THW vapour is circulated through a desiccant drier system to recover the heavy water. The driers operate with continuous cycles of an ad-

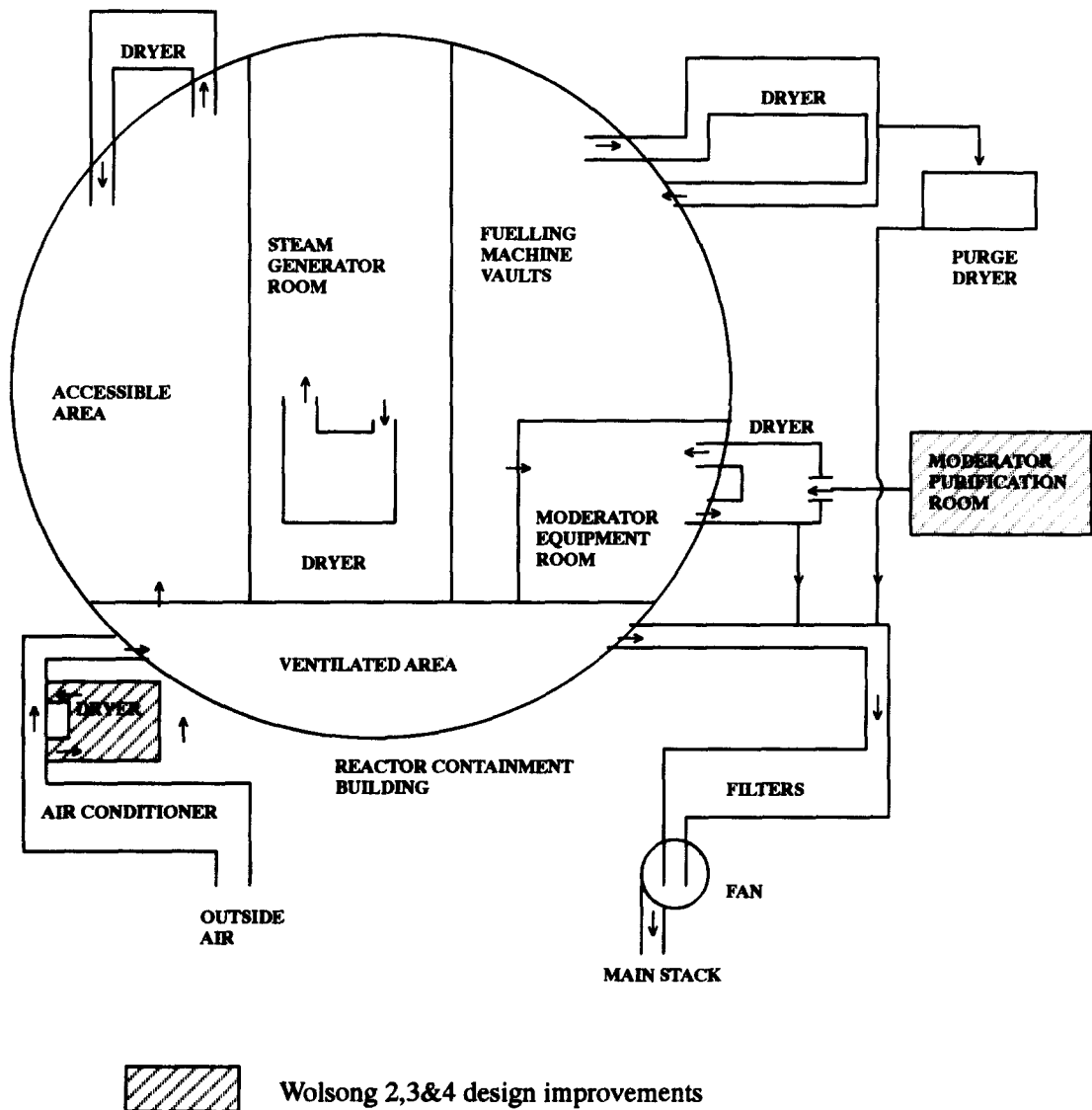


Fig. 4. A schematic diagram of the ventilated and dried areas in the reactor building at CANDU 6 GS.

sorption mode followed by a regeneration mode. During the regeneration mode, the THW is removed as condensate.

Airborne tritium is displaced by passing clean air through the contaminated area. The air is supplied from outside the area of containment by the ventilation system. For the Wolsong 2, 3 and 4 design, additional driers are installed (see Fig. 4) at the air intake filter to the reactor building

ventilation system. The additional driers will reduce the humidity of the ventilation air entering the reactor building. Thus the condensate collected in the D_2O vapour recovery driers will have a higher isotopic value.

Tritium confinement is achieved by a suitable layout of the system and by appropriate atmospheric separation of areas which are prone to high specific activity THW leakage.

In spite of the tritium control features on the plant, a small amount of THW is discharged to the environment via the main stack.

Liquid THW is collected and recovered via active drainage systems. If a spill has occurred in the reactor building, most of the THW is recovered by operators using portable D_2O recovery tools. Unrecovered THW passes through floor drains by gravity and is then collected in the eight sumps and pumped to the active drainage tank. The collected THW is transported to the D_2O clean-up system or the liquid radioactive waste management system, depending on the isotopic value of D_2O . Considering the station economics, D_2O with an isotopic value of greater than 0.5‰ is transported to the D_2O clean-up system; that with a value of less than 0.5‰ is transported to the liquid radioactive waste management system.

Unrecovered THW in the service building is collected via the active and low active drainage systems into the active sump and low active sump respectively. These are located in the service building basement. The low activity waste, which includes tritium collected in the low active sump, is pumped into one of three low active liquid waste tanks. Active waste, which includes tritium collected in the active sump, is pumped into one of the two active liquid waste tanks. Waste from special sources (e.g. reactor building drains, heavy water areas, resin storage area, spent fuel bay, spent fuel bay underdrainage ground water sump and reactor building and service building underdrainage ground water sump) is normally treated as active waste.

A decontamination facility comprising a filter and an ion exchange column is provided to process large quantities of liquid radioactive effluent prior to discharge into the environment via the condenser cooling water duct. The decontamination facility is not effective for waterborne tritium, but is effective for ionic radioisotopes.

6. Discussion

The major tritium sources of the CANDU 6 GS are the moderator and the coolant systems. The tritium build-up rate in the moderator system is

greater than that in the coolant system. This can be explained as follows: a larger fraction of the moderator inventory is exposed to the neutron flux inside the calandria at any given time, i.e. approximately 91% versus 5% for the coolant system. The neutron flux level and the activation cross-section for the $D(n, \gamma)T$ reaction are greater in the moderator D_2O . The generation and total inventory of tritium in the shield cooling system and the liquid zone control system are minor compared with those in the moderator system and the coolant system (see Table 1).

From Wolsong 1 experience, shown in Table 2, the increase in tritium activity for the moderator system is well below that of the theoretical value which is based on operation at 84% capacity factor except for the year of 1990. Conversely, the tritium activity of the coolant system has been higher than the theoretical value since 1991. It is believed that the ingress of airborne tritium from the moderator enclosure into coolant related dried areas has caused the increase in tritium activity in the primary coolant system. A cross-contamination from the moderator system to the coolant system causes the tritium activity to increase in the coolant heavy water. As a result, the radiation exposure to station personnel will be increased.

Leakage from a pressurized heavy water system occurs at mechanical joints. Stem sealed gate valves and flanges are considered to be major contributors. The threshold detectable leak rate of heavy water is 1 g day^{-1} (Isles et al., 1980). Thousands of 1 g day^{-1} leaks add up to an appreciable tritium hazard and yet each individual leak is too small to be worth stopping when the equipment is installed and the plant is operating. The solution is to identify how each kind of device tends to leak in service and to prevent these leaks through various measures in the design, specifications, manufacture and installation of new equipment.

Even though individual leakage of THW can be prevented, it is prudent to assume that there will continue to be some leakage of tritium. In the absence of the leakage mentioned above, spills of THW can occur during transfer processes and when process equipment is opened for maintenance, etc.

Tritium in vapour form can spread to adjacent areas. Two different systems, the ventilation system and D₂O vapour recovery system, are provided to reduce tritium concentration in air in the reactor building. Each 50% reduction in the tritium field requires a doubling of the ventilation flow rate (Isles et al., 1980). Confinement (by installing one or more physical barriers surrounding each area where tritium leaks to the air) and vapour recovery are the preferred design countermeasures to reduce airborne tritium concentration.

The tritium releases for a plant are estimated from heavy water management data for the coolant and moderator systems. Based on Wolsong 1 experience, THW losses from the coolant system constitute 94% (on average) of the total losses of heavy water (Park and Aydogdu, 1993). These losses from the coolant represent about 35% of the tritium releases into the environment. In contrast, the moderator system contributes less than 6% of the total losses of heavy water, but due to its higher tritium activity, this constitutes 65% of the tritium releases.

7. Conclusions

There are several tritium generating reactions for the CANDU 6 GS. The major mechanism for tritium generation is the D(n, γ)T reaction in the moderator and coolant systems. The theoretical equilibrium tritium activity level in the moderator, based on operation at an 84% capacity factor, is 3.26×10^{12} Bq kg-D₂O⁻¹ and that of the coolant is 7.21×10^{10} Bq kg-D₂O⁻¹.

As the operating time increases, the ingress of airborne tritium from the moderator enclosure into the coolant related dried areas causes the tritium activity to increase in the coolant system. To reduce intermixing between the moderator and coolant heavy water, confinement and strict heavy water management are important controlling factors during the design stage and operation stage respectively.

The prevention of leakage from the THW filled systems has a high priority during the design stage and must have an equally high priority during the operation stage. Thus, to reduce leakage from the THW filled systems, unnecessary mechanical joints and components should be eliminated. Examples include the replacement of flanged joints with welded joints and the removal of any unnecessary valves. Also, plant operating personnel should always place a priority on preventative maintenance activities during plant operation.

The driving force of tritium transport in vaporized THW is ventilation and D₂O vapour recovery air flow. Vaporized THW is confined and recovered via separately installed D₂O vapour recovery drier units in each area. The driving force of tritium transport in pathways aqueous THW is gravity and pumping power. Active THW in the reactor building is collected via the active drainage system.

The CANDU 6 GS has unique THW collection and recovery systems. As a result, tritium releases to the environment are small, and the evaluated dose to a member of the public critical group has been only approximately 0.1% of the ICRP-26 regulatory limit of 5000 μ Sv year⁻¹. With design improvements in the pathways of tritium transport, it is expected that the recovery of THW will be increased, and tritium releases to the environment and the resulting public dose will be even lower for Wolsong 2, 3 and 4.

References

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